Reg. No. | | | | | | | | | | | [18MTDS101]

FIRST SEMESTER M.TECH EXAMINATION, JANUARY 2019

DISCRETE MATHEMATICS AND LINEAR ALGEBRA

Time: 03 Hours **Maximum Marks:100**

Instructions :
- **i. Answer Five full questions. Question No. 1 and 2 are Compulsory.**
- **ii. Draw neat sketches wherever necessary**
- **iii. Missing data may be assumed suitably.**

1. a. Define a function and types of function with suitable examples. **07**

b. (i) Find the truth value of each proposition if p and r are true and q is False. (a) $\neg p \wedge (q \wedge r)$ (b) $p \wedge (\neg(q \wedge r))$ **07**

(ii) Prove that for any propositions p, q, r the compound proposition $[(p \vee q) \wedge \{(p \to r) \wedge (q \to r)\}] \to r$ Is a tautology.

c. Define Quantifiers. Negate and simplify each of the following **06**

(a) $\exists x, [p(x) \vee q(x)]$ (b) $\forall x, [p(x) \wedge \neg q(x)]$

(c) $\forall x, [p(x) \to q(x)]$ (d) $\exists x, [\{p(x) \vee q(x)\} \to r(x)]$

2. a. Find the number of permutations of the letters of the word MASSASAUGA, In how many these all four A's are together? How many of them begin with S? **07**

b. How many integers between 1 and 300 (inclusive) are **07**

i. Divisible by at least one of 5, 6,8 ?

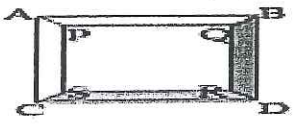
ii. Divisible by none of 5, 6,8 ?

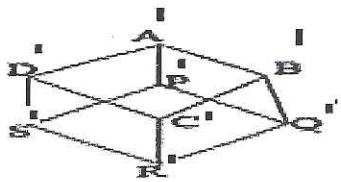
c. Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n \ for \ n \geq 0 \ given \ F_0 = 0, F_1 = 1$ **06**

3. a. Define the following terms with an example for each **06**

(i) Graph (ii) Degree of a vertex (iii) Sub graph (iv) Spanning sub graph

b. Prove that following graphs are isomorphic **07**





c. Explain the KONIGSBERG BRIDGE problem. **07**

OR

4 a. Define the following terms with an example for each **06**

(i) Bipartite graph (ii) Path (iii) Euler circuit (iv) Euler graph

b. **07**

Use Warshall's algorithm for A={1,2,3,4} and w= $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

c. Define chromatic Number. Find the chromatic polynomial for the cycle c_4 of the length 4. What is its chromatic number? **07**

5. a. Show that the transformation $y_1 = 2x_1 - 2x_2 - x_3$, **06**

$y_2 = -4x_1 + 5x_2 + 3x_3$ $y_3 = x_1 - x_2 - x_3$ is regular and find its transformation.

b. **07**

Using elementary transformation find the rank of matrix A= $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

c. Explain Leontief model. An economy is based on three sectors, agriculture (A), energy (E) and manufacturing (M). Production of a dollar's worth of agriculture requires an input of $\$0.2$ from the agriculture sector and $\$0.4$ from the energy sector. Production of a dollar's worth of energy requires an input of $\$0.2$ from the energy sector and $\$0.4$ from the manufacturing sector. Production of a dollar's worth of manufacturing requires an input of $\$0.1$ from the agriculture sector, $\$0.1$ from the energy sector, and $\$0.3$ from the manufacturing sector. Find the output from each sector that is needed to satisfy a final demand of $\$20$ billion for agriculture, $\$10$ billion for energy, and $\$30$ billion for manufacturing. **07**

OR

6. a. Show that $\{u_1, u_2, u_3\}$ is an orthogonal set where **07**

$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$ express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as linear combination of vector S.

b. Use Gram–Schmidt orthonormalization. $v_1=\begin{bmatrix}1\\1\\1\\1\end{bmatrix}, v_2=\begin{bmatrix}0\\1\\1\\1\end{bmatrix}, v_3=\begin{bmatrix}0\\0\\1\\1\end{bmatrix}$ to construct the orthogonal basis for w. **07**

c. Explain the application of Orthogonal diagonalisation of symmetric matrices **06**

7. a. Verify Cayley-Hamilton theorem **06**

$$A=\begin{bmatrix}1 & 1 & 3\\1 & 3 & -3\\-2 & -4 & -4\end{bmatrix}$$

b. Reduce the matrix to the diagonal form **07**

$$A=\begin{bmatrix}8 & -6 & 2\\-6 & 7 & -4\\2 & -4 & 3\end{bmatrix}$$

c. Solve the system $X'=\begin{bmatrix}-5 & 1\\4 & -2\end{bmatrix}X$. Also discuss the stability of the system **07**

OR

8. a. Verify that $T:V_3(R)\to V_3(R)$ defines as $T(x,y,z)=(x-y+2z, 2x+y-z, -x-2y)$ is linear transformation **06**

b. Diagonalize the symmetric matrix A. **07**

$$A=\begin{bmatrix}3 & 2 & 0\\2 & 2 & 2\\0 & 2 & 1\end{bmatrix}$$

c. If $q(x)=4x^2+3y^2+z^2-8xy-6yz+4xz$ find the rank, index and nature of quadratic form. **07**